

SDM234-Homework2

1. For a complex function f , prove that:

(a) If $\lim_{z \rightarrow z_0} f(z)$ exists, then the limit is unique.

(b) Assume f is continuous. If there exists a sequence of complex numbers $z_1, z_2, \dots$ such that

$$|z_n - z_0| < \frac{1}{n}, \quad n = 1, 2, \dots,$$

then

$$\lim_{n \rightarrow \infty} f(z_n) = f(z_0).$$

(c) The function f has at most one derivative value. (Hint: Use part (a) and the definition of the derivative.)

2. Find out, and give a reason, whether f is continuous at $z = 0$ if $f(0) = 0$ and, for $z \neq 0$, the function is equal to:

(1) $\frac{1}{2i} \left(\frac{z}{\bar{z}} - \frac{\bar{z}}{z} \right);$

(2) $\frac{\operatorname{Re}(z^2)}{1 - |z|}.$

3. Calculate the derivatives of:

(1) $\frac{1}{z^2 + 2};$

(2) $(z - 1)^2(z^2 + 3).$

4. Use the Cauchy–Riemann equations to analyze the differentiability and analyticity of the following functions, where $z = x + iy$:

(1) $f(z) = \sin x \cosh y + i \cos x \sinh y;$

(2) $f(z) = x^3 - y^3 + 2ix^2y^2.$

5. Determine and graph the following sets in the complex plane:

(1) $|z - 2i| \leq 1;$

(2) $\left| \frac{1}{z} \right| < 2;$

(3) $\arg(z - i) = \frac{\pi}{4}.$